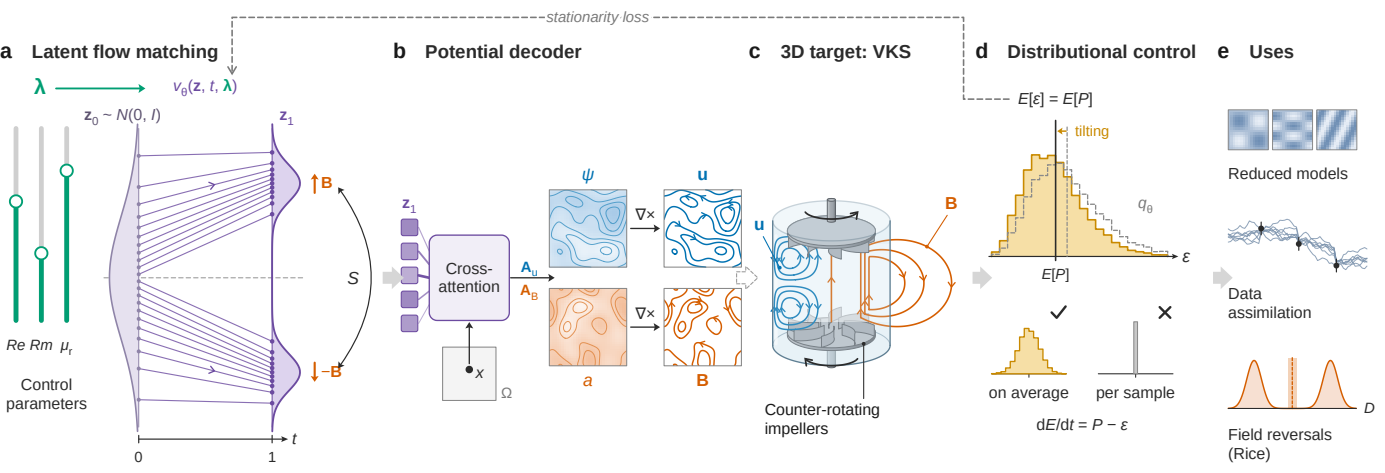

stationarity loss
a Latent flow matching
λ
$v_\theta(\mathbf{z}, t, \boldsymbol{\lambda})$
$\mathbf{z}_0 \sim N(0, I)$
$\mathbf{z}_1$
B
S
−B
t
0
1
Re Rm μ_r
Control parameters
b Potential decoder
$\mathbf{z}_1$
Cross-attention
A_u
A_B
x
Ω
ψ
u
∇×
a
B
c 3D target: VKS
u
B
Counter-rotating impellers
d Distributional control
$E[\varepsilon] = E[P]$
tilting
q_θ
ε
$E[P]$
on average
per sample
$dE/dt = P - \varepsilon$
e Uses
Reduced models
Data assimilation
Field reversals (Rice)
D
C3 symmetry B ↦ −B: equivariance
C1 ∇·u = ∇·B = 0: by construction
tested: 3-mode, 2D MHD
C4 mean energy budget